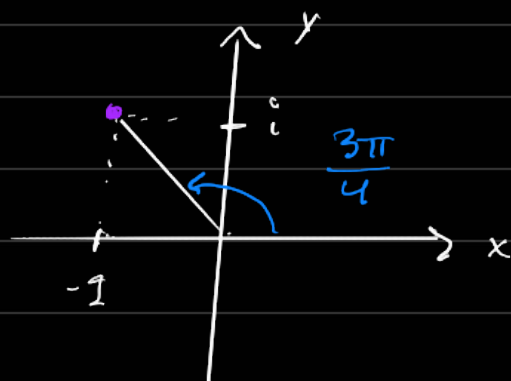


Short extra: Use $e^{i\theta} = \cos(\theta) + i\sin(\theta)$

- Note $(-1 + i)$ is:
which has $\theta = \frac{3\pi}{4}$, thus



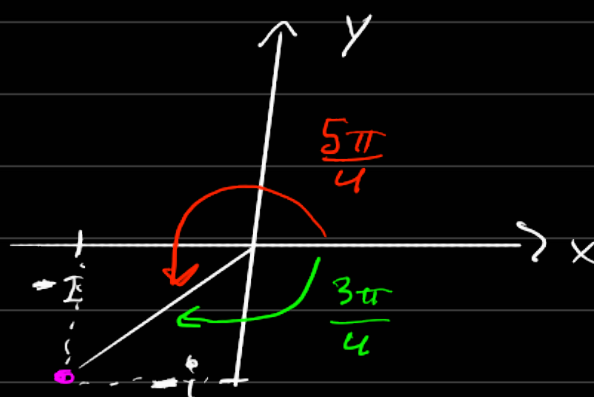
$$e^{i\frac{3\pi}{4}} = \underbrace{\cos\left(\frac{3\pi}{4}\right)}_{-1} + i \underbrace{\sin\left(\frac{3\pi}{4}\right)}_{+1}$$

$$e^{i\frac{3\pi}{4}} = -1 + i \quad \textcircled{a}$$

- likewise: $(-1 - i)$:

$$e^{i\frac{5\pi}{4}} = \underbrace{\cos\left(\frac{5\pi}{4}\right)}_{-1} + i \underbrace{\sin\left(\frac{5\pi}{4}\right)}_{-1}$$

$$\therefore e^{i\frac{5\pi}{4}} = -1 - i$$



Now, from $e^{i\theta} = \cos(\theta) + i\sin(\theta)$ you can easily find that:

$$\sin(\theta) = \frac{e^{i\theta} - e^{-i\theta}}{2i} \quad \textcircled{b}$$

→ Final thing: $e^{i\frac{5\pi}{4}} = e^{-i\frac{3\pi}{4}} \quad \textcircled{c} \quad (\text{see above})$

* Use $\textcircled{a}$, $\textcircled{b}$, $\textcircled{c}$ to get to the answer they give you after using Cauchy's Res. Thm.

